

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

QNS

GRAVITATION

LECTURE-4

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1** Revision
- 2** Gabbar Numerical Series PYQs
- 3** ..
- 4** ..
- 5** ..

Recap From 1st Lecture

* Universal Law of Gravitation

$$F = G \frac{m_1 m_2}{r^2} \quad (m_1) \text{---} r \text{---} (m_2)$$

$$G = 6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

Gravitational

Vector form of ~~Coulomb's~~ Law

$$F = \frac{G m_1 m_2}{(r_2 - r_1)^3} (\vec{r}_2 - \vec{r}_1)$$

JKBOSE

Acceleration due to Gravity



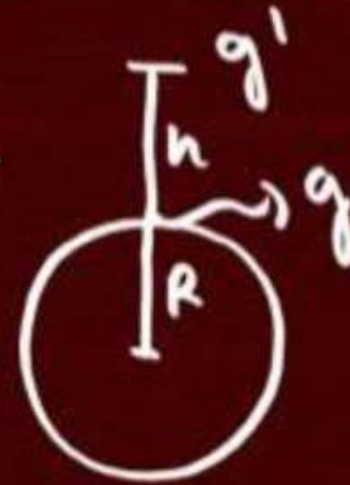
$$g = \frac{G M}{r^2}$$

$$g = G \frac{4}{3} \pi \rho r$$

$$g = -10 \text{ m/s}^2$$

$$g = -9.8 \text{ m/s}^2$$

Variation of g with height

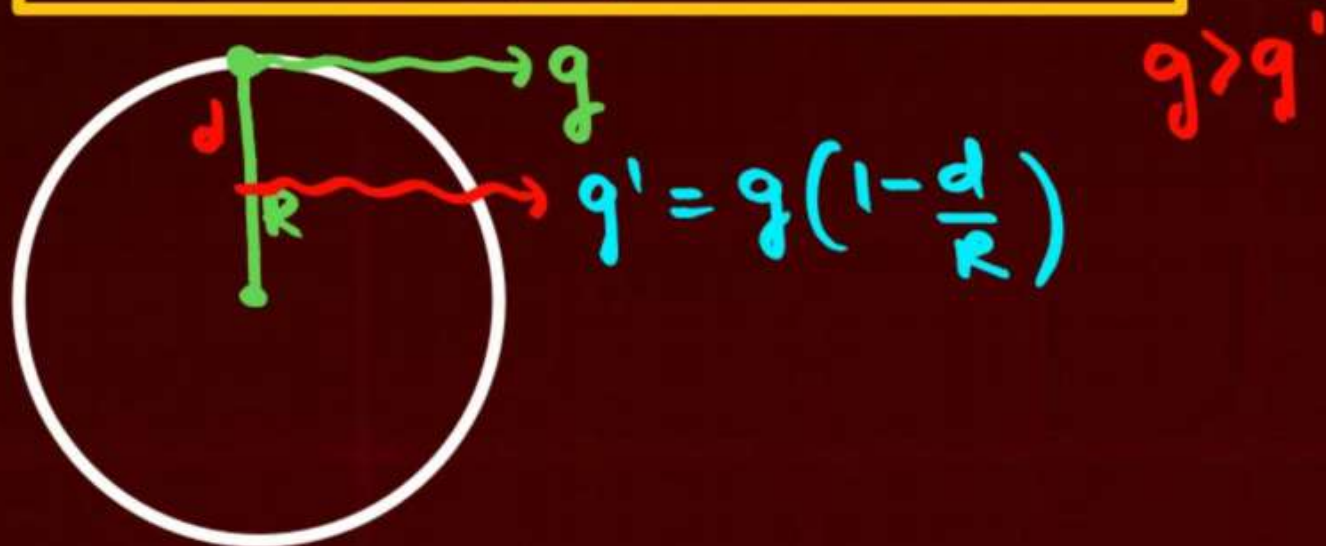


$$g' = g \left(1 - \frac{2h}{R} \right)$$

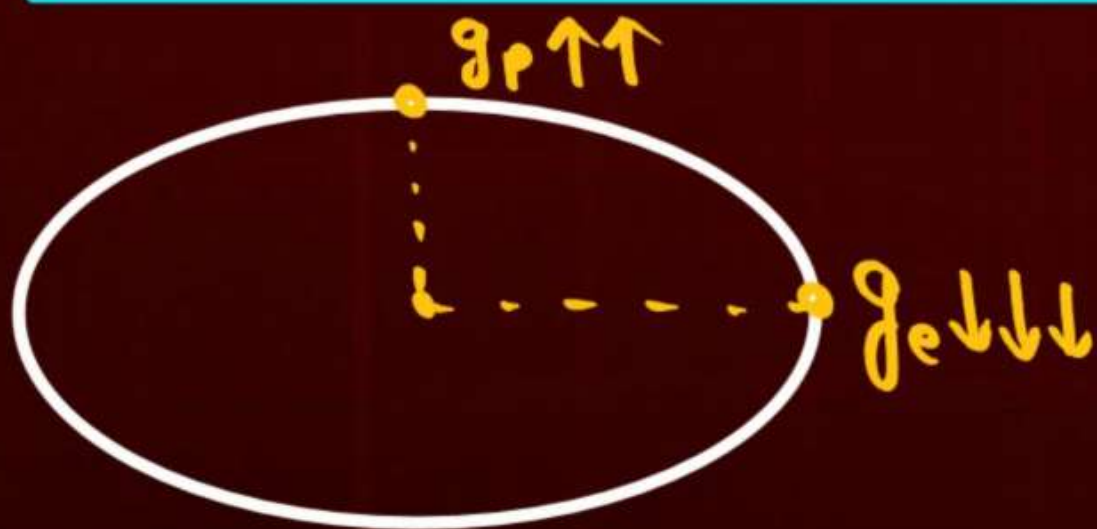
$$g > g' \checkmark$$

Recap From 2nd Lecture

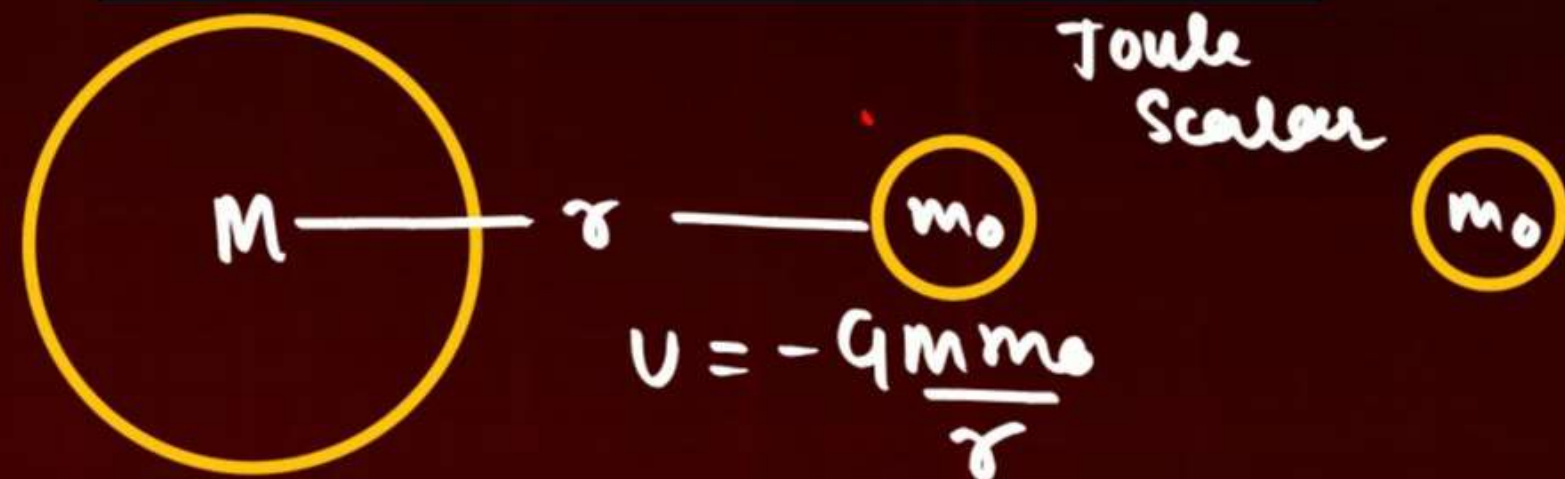
Variation of g with depth



Variation of g due to Shape of Earth



Gravitation Potential Energy



Gravitational potential

✓ $V = \frac{U}{m_0}$ J/kg Scalar

✓ $V = -G \frac{M}{r}$

Recap From 3rd Lecture

• Escape Velocity



$$v_e = \sqrt{\frac{2Gm}{r}}$$

$$v_e = \sqrt{2rg}$$

$$v_e = 11.2 \text{ Km/sec}$$

Orbital velocity



$$v_o = \sqrt{\frac{Gm}{r}}$$

$$v_o = \sqrt{rg}$$

$$v_o = 7.9 \text{ Km/sec}$$

Relationship b/w v_e and v_o

$$v_e = \sqrt{2} v_o$$

Kepler law of Planetary Motion

1 law of orbit



2 law of Area

$$A_1 = A_2$$

law of T

$$T^2 \propto a^3$$

Lecture.no.4

Gabbbar

Numerical Series

PYQs

QUESTION [JKBOSE SUMMER SET-A 2026]

Q Which law describes the orbits of planets around the sun?

- A** Newton's law
- B** Faraday's law
- C** ✓ Kepler's law
- D** Kirchhoff's law

QUESTION [JKBOSE SUMMER SET-A 2026]

The gravitational force between two blocks is F . What would happen if the mass of both the blocks as well as distance between them is doubled?

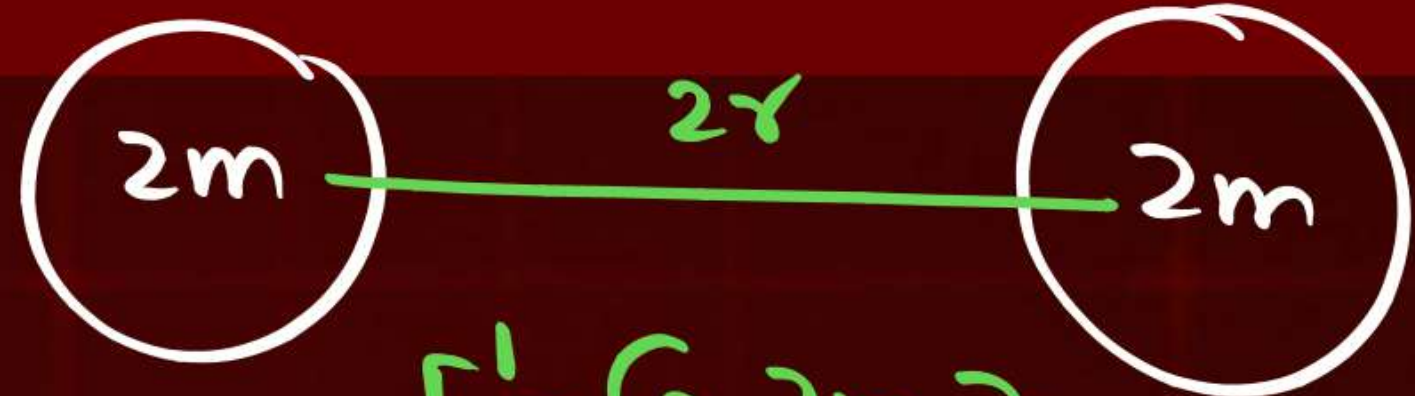
Sol //



$$F = G \frac{mm}{r^2}$$

$$F = F'$$

no effect.



$$F' = G \frac{2m \cdot 2m}{(2r)^2}$$

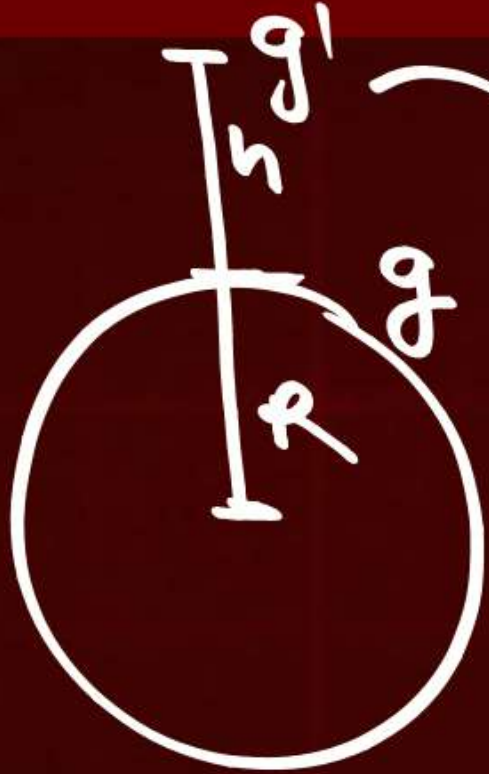
$$F' = G \frac{4mm}{4r^2}$$

$$F' = G \frac{mm}{r^2}$$

QUESTION [JKBOSE SUMMER SET-A 2026]

Explain how g varies with altitude.

- 3m ✓



$$g' = g \left(1 - \frac{2h}{R}\right)$$

QUESTION [JKBOSE SUMMER SET-B 2026]

The formula for Kepler's third law is:

$$\underline{T^2 \propto a^3}$$

A $T^3 \propto a^2$

B $T^3 \propto a^3$

C $T^2 \propto a^2$

D $T^2 \propto a^3$

QUESTION [JKBOSE SUMMER SET-C 2026]

Kepler's second law of planetary motion is known as:

- A** The law of orbits
- B** The law of areas ✓
- C** The law of periods
- D** The law of gravity

QUESTION [WINTER ZONE SET-A 2025]

(Neet-2022)

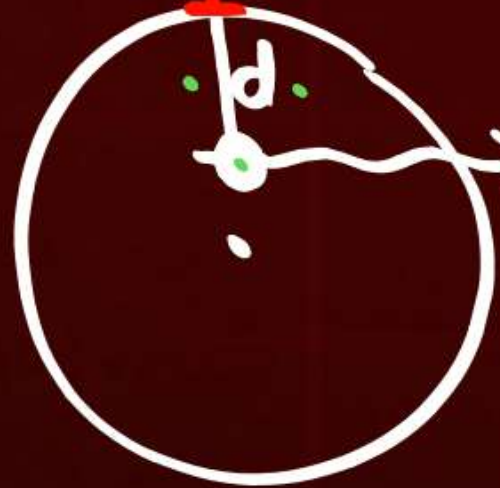


At what height above earth's surface, value of " g " is same as in a 100 km deep mine? 3m

Sol

$$d = 100 \text{ km}$$

$$g' = g \left(1 - \frac{2h}{R}\right)$$



$$g' = g \left(1 - \frac{d}{R}\right)$$

$$g_h = g_d$$

$$g \left(1 - \frac{2h}{R}\right) = g \left(1 - \frac{d}{R}\right)$$

$$1 - \frac{2h}{R} = 1 - \frac{d}{R}$$

$$1 - 1 - \frac{2h}{R} = -\frac{d}{R}$$

$$\frac{2h}{R} = \frac{d}{R}$$

$$2h = d$$

→ 100 km

$$h = \frac{d}{2}$$

$$h = \frac{100}{2}$$

$$h = \underline{50 \text{ km}}$$

QUESTION [WINTER ZONE SET-A 2025] ✓

Find an expression for the gravitational potential energy of a body situated at a distance " r " from the centre of earth. 3m

QUESTION [SUMMER ZONE SET-A 2025]

✓ At what depth below the surface of earth the value of ' g ' is same as that of height of 5 km?

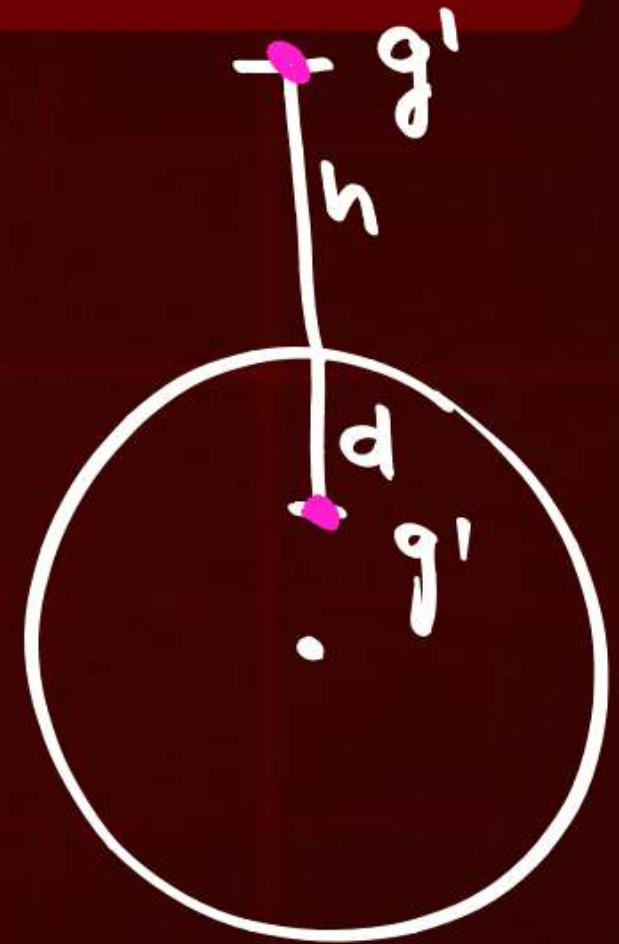
- A** ✓ 10 km
- B** 7.5 km
- C** 5 km
- D** 2.5 km

$$h = 5 \text{ km}$$

$$d = 2h$$

$$d = 2h$$

$$d = 2 \times 5 = 10 \text{ km}$$



QUESTION [SUMMER ZONE SET-A 2025]

✓ State universal law of gravitation. Derive relation between ' g ' and ' G '.

$$F \propto m_1 m_2$$

$$F \propto \frac{1}{r^2}$$

$$F \propto \frac{m_1 m_2}{r^2}$$

$$F = G \frac{m_1 m_2}{r^2}$$

$g \Rightarrow$ acceleration due to gravity
 $G \Rightarrow$ Gravitation constant

$$g = \frac{GM}{r^2}$$

QUESTION [SUMMER ZONE SET-A 2025]

Define orbital velocity. Derive an expression for it.

$$V_0 = \sqrt{\frac{Gm}{r}}$$

$$V_0 = \sqrt{rg}$$

$$V_0 = 7.9 \text{ km/sec}$$

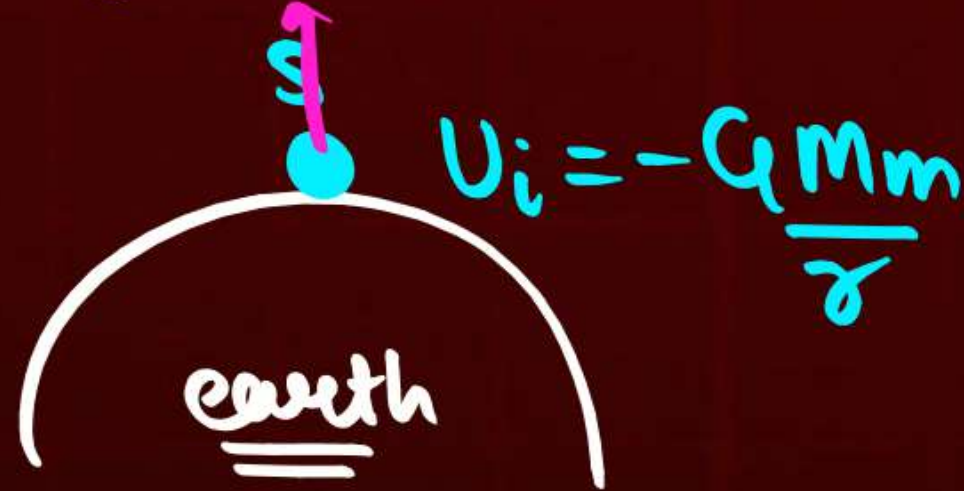
QUESTION [SUMMER ZONE SET-B 2025]

When a missile is launched with a velocity less than the escape velocity, the sum of kinetic and potential energy is:

- A** Zero
- B** Negative
- C** Positive
- D** Infinite

12th $\Rightarrow$ Sto. of Atom & nuclei

$v_e > v$ $K.E = \frac{1}{2}mv^2 = 8$



$v_e = 0$ $K.E = 0$

$U_i = -10$

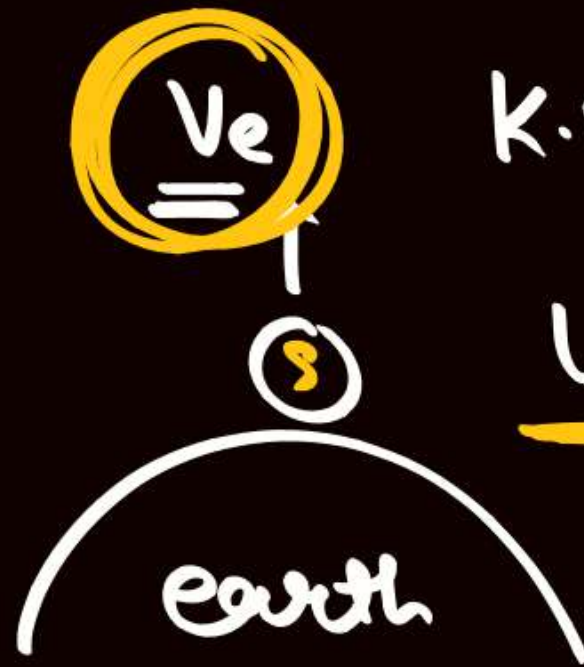
$T.E = U_i + K_i$
 $= -10$

$v \neq 0$ $K.E_i = 8$

$U_i = -10$

$T.E = U_i + K_i$
 $= -10 + 8 = -2$

#



$$K.E = \frac{1}{2}mv_e^2 = 12J$$

$$U_i = -10$$

$$\begin{aligned} T.E &= K.E_i + U_i \\ &= 12 + -10 \\ &= 12 - 10 \\ &= +2J \end{aligned}$$

QUESTION [SUMMER ZONE SET-B 2025]

State universal law of gravitation. Derive relation between 'g' and 'G'.

QUESTION [SUMMER ZONE SET-C 2025]

Which law describes the orbits of planets around the sun?

- A** Newton's law
- B** Kirchhoff's law
- C** Faraday's law
- D** Keplar's law

QUESTION [JKBOSE SET-X 2024]

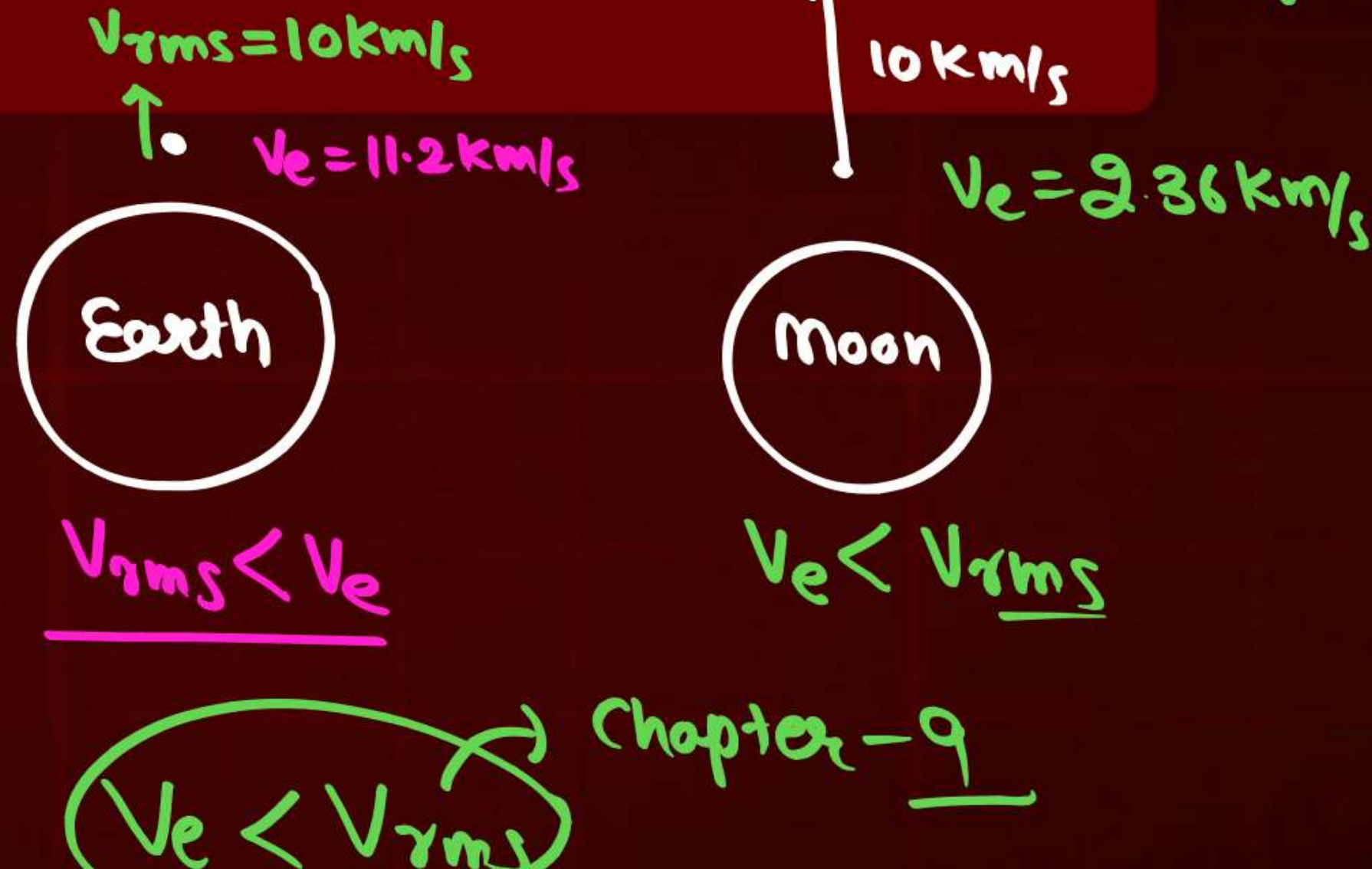
RMS → Root mean Square Velocity



There is no atmosphere on moon as:

(Neet-2019)

- A** It is closer to earth
- B** It revolves around the earth
- C** It gets light from the sun
- D** The escape velocity of gas molecules is less than their r.m.s. velocity on moon



QUESTION [JKBOSE SET-X 2024]

State Kepler's law of planetary motion (any two). ✓

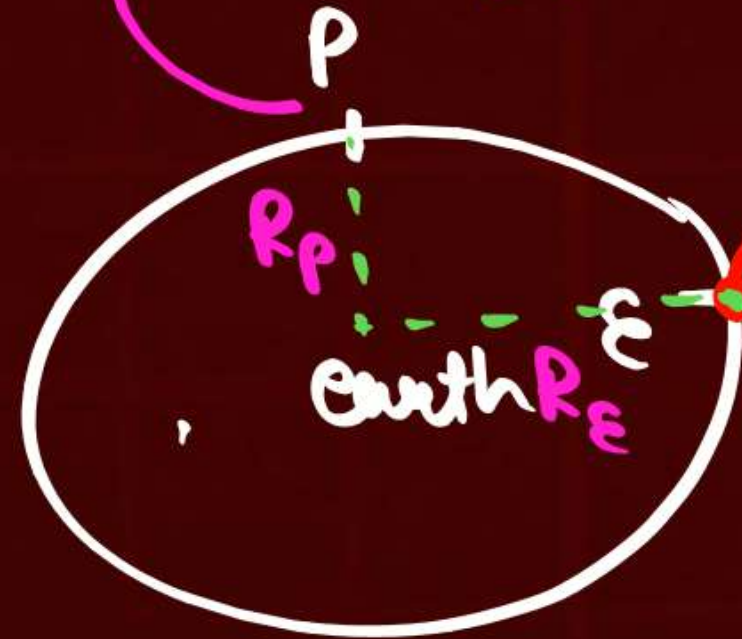
QUESTION [JKBOSE SET-X 2024]

Define escape velocity. Derive an expression for it. ✓

QUESTION [JKBOSE SET-Y 2024]

Where will it be profitable to purchase 1 kg sugar?

$$Rs = 100/- \quad \uparrow \underline{W_p = mg_p} \quad \uparrow \quad g_p = \frac{Gm}{R_p^2} \downarrow$$



$$\begin{aligned} & \uparrow R_e \quad g_e \downarrow \downarrow \\ & \downarrow \underline{W_e = mg_e} \\ & Rs = 50/- \end{aligned}$$

A At poles

B At equators

C At 45° latitude

D At 40° latitude

QUESTION [JKBOSE SET-Z 2024]

(T)

The time period of an earth satellite in circular orbit is independent of:

- ☒ **A** The mass of the satellite
- ☒ **B** Radius of its orbit
- ☒ **C** Both ~~mass~~ and radius of orbit
- ☒ **D** Neither mass of satellite nor the radius of its orbit



$$V_0 = \sqrt{\frac{Gm}{r^2}}$$

$$\left(\frac{2\pi r}{T} \right) = \sqrt{\frac{Gm}{r^2}}$$

$$\frac{T}{2\pi r} = \sqrt{\frac{r^2}{Gm}}$$

$$\textcircled{T} = 2\pi r \sqrt{\frac{r^2}{Gm}}$$

$$V_0 = \frac{2\pi r}{T}$$

SUMMARY



Qabber numerical series

HOMework



Write all Question
On note book

“Insaan nahi bolta uske din
bolate hai or jab din bolte hai to
behre bhi sunte hain”

Join Our Official Telegram & Whatsapp Channel



JKBOSE Wallah
Class 9th & 10th



JKBOSE Wallah



Scan the QR to Join

Thank
You

